CREW CHANGE
RISK COPILOT
AI-SUPPORTED OPERATIONAL INTELLIGENCE

AI for Smarter Crew Change Operations

Capstone Poject - IronHack - AI Consilting and Intergration - 2026-09-01

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AI for Smarter Crew Change Operations

Exploring AI opportunities for a medium-sized ship management company

THE CHALLENGE

⚓ **Vessel schedules**

👥 **Crew availability**

✈️ **Travel & airport disruption**

🌍 **Port & operational constraints**

CHLEO'S QUESTION

"What is the AI actually doing, and can we trust it?"

AI OPPORTUNITIES EXPLORED

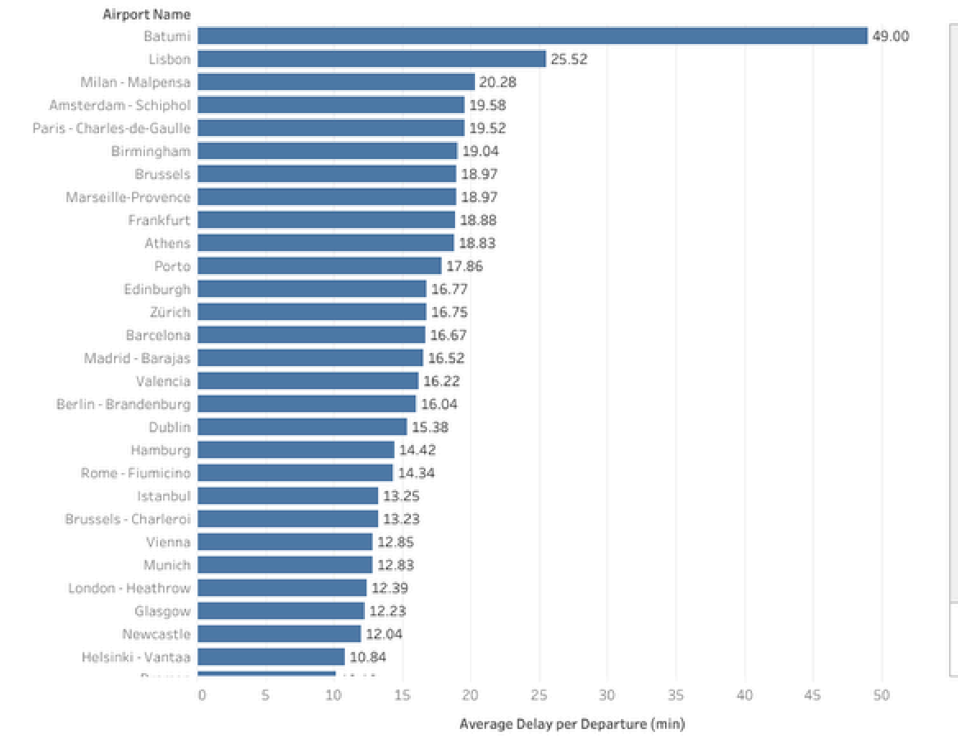
1. **Crew Change Risk Copilot ★**
2. **AI Operational Briefing & Exception Reporting**
3. **AI-Assisted Communication & Document Analysis**

Focus: practical operational value while keeping human judgement and accountability.

Operational Context: Where Could Disruption Occur?

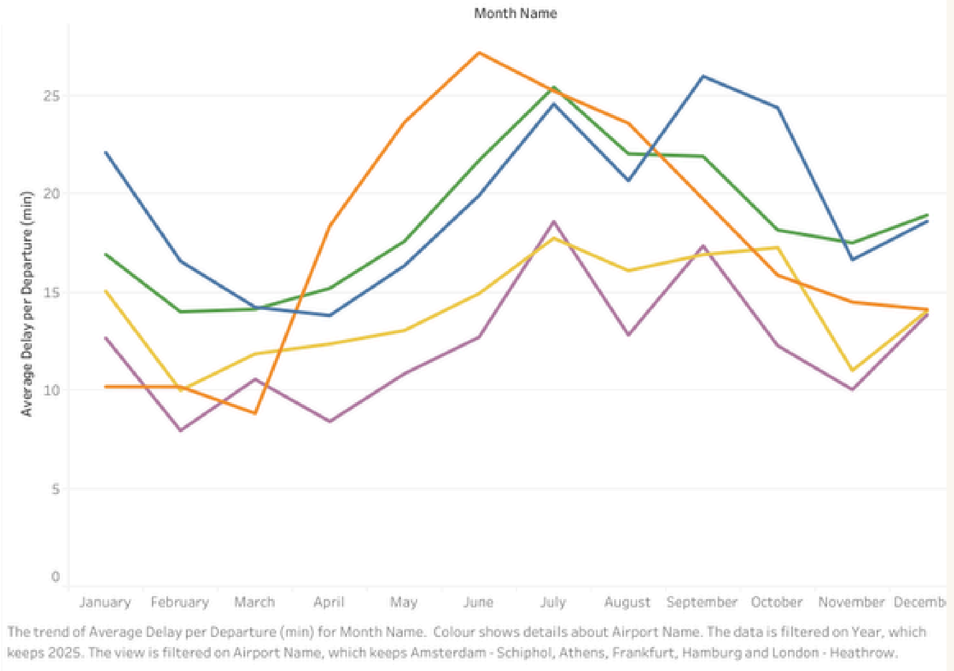
Using available data to understand workforce and travel-related operational factors

European Airport Delay Ranking 2025



Average Delay per Departure (min) for each Airport Name. The marks are labelled by Average Delay per Departure (min). The data is filtered on Year, which keeps 2025. The view is filtered on Airport Name, which keeps 32 of 103 members.

Seasonality of Average Pre-Departure Delays — 2025

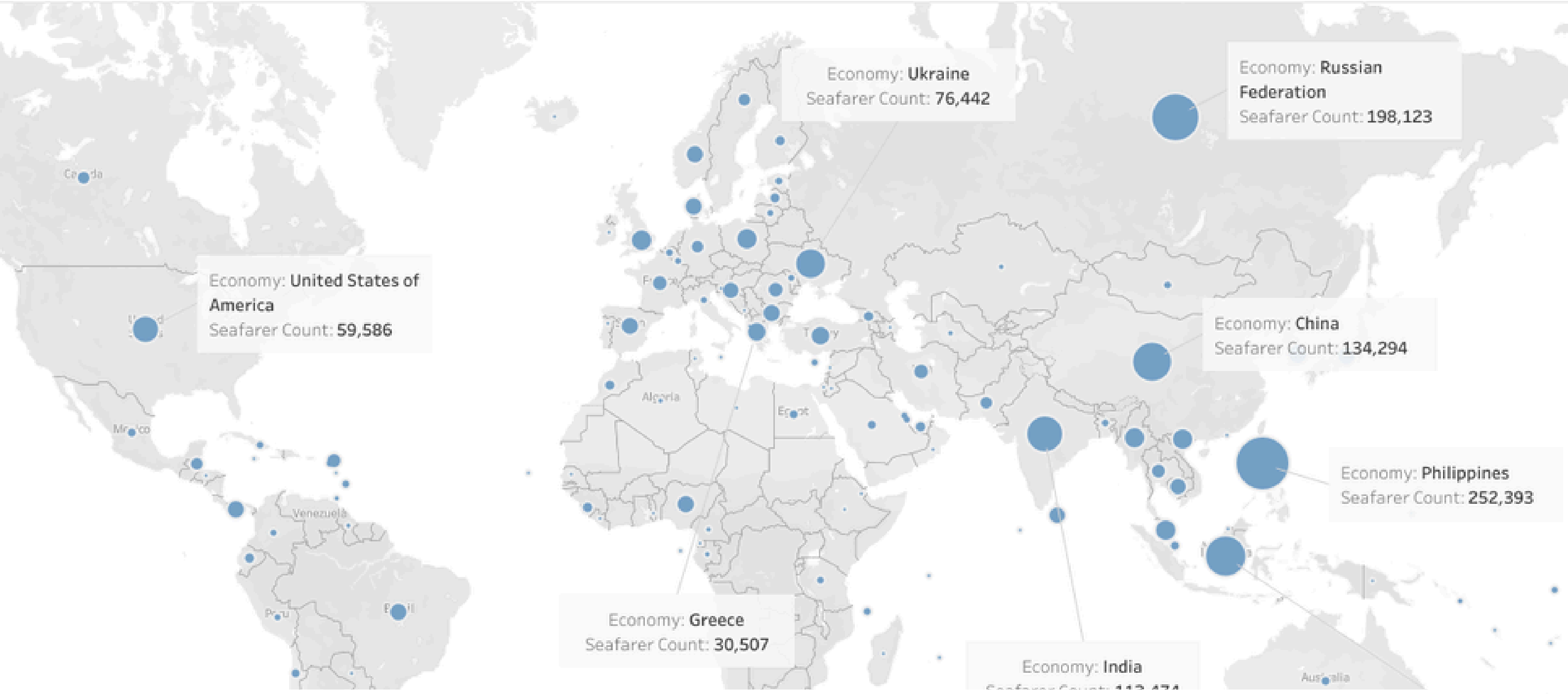


The trend of Average Delay per Departure (min) for Month Name. Colour shows details about Airport Name. The data is filtered on Year, which keeps 2025. The view is filtered on Airport Name, which keeps Amsterdam - Schiphol, Athens, Frankfurt, Hamburg and London - Heathrow.

Airport Name
Amsterdam - Sc.. | Frankfurt | London - Heathr..
Athens | Hamburg

SOURCE: EUROCONTROL | Operational air traffic data | Analysis year: 2025

Total Seafarer Supplier Countries 2021

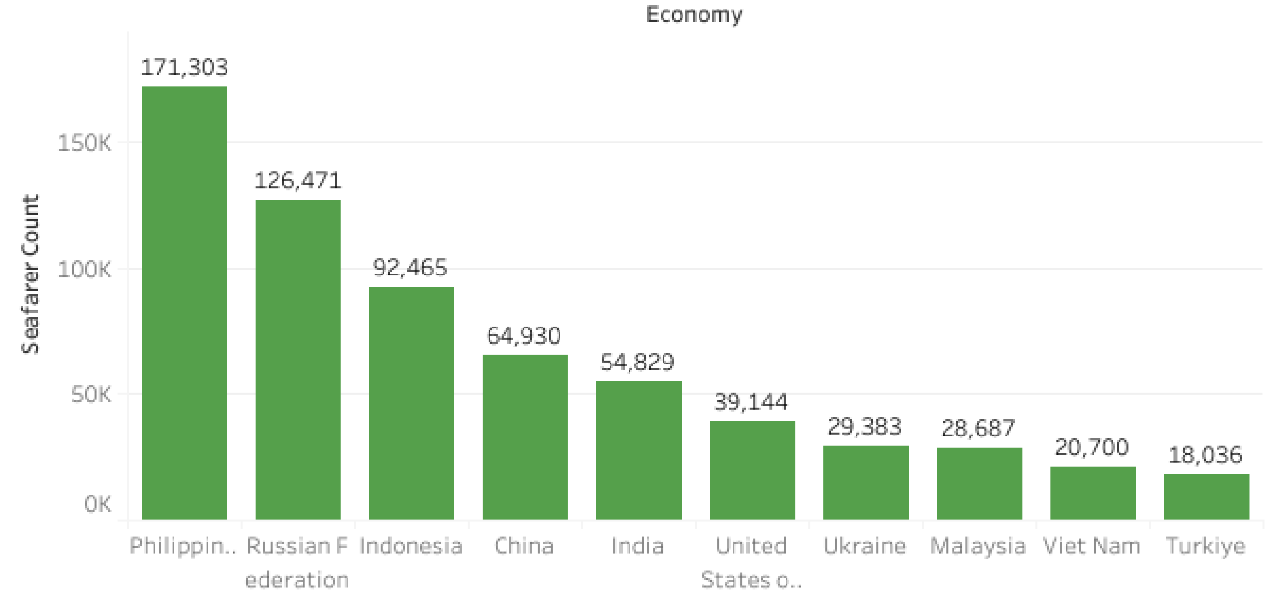


SOURCE: UNCTADstat (US.Seafarers) | Estimated seafarer supply by economy and rank | Data years: 2015 & 2021

WHY THIS MATTERS FOR CREW CHANGE OPERATIONS

- Crew changes depend on multiple interconnected factors
- Travel disruptions can affect crew mobilisation and repatriation
- Workforce availability varies across different crew source markets
- Operations teams must combine information from multiple sources
- Opportunity for AI-supported monitoring and prioritisation

Top Seafarer Suppliers — Ratings 2021



Crew Change Risk Copilot

AI-supported decision support for proactive identification and review of potential crew-change disruptions

OPERATIONAL INPUTS

- **Crew change plans**
- **Travel information**
- **External context**

→

CREW CHANGE RISK COPILOT

- **Monitors**
- **Identifies**
- **Summarises**
- **Explains**

→

HUMAN REVIEW

- **Prioritise cases**
- **Assess context**
- **Make decisions**

IDENTIFY

Potential cases requiring additional attention

EXPLAIN

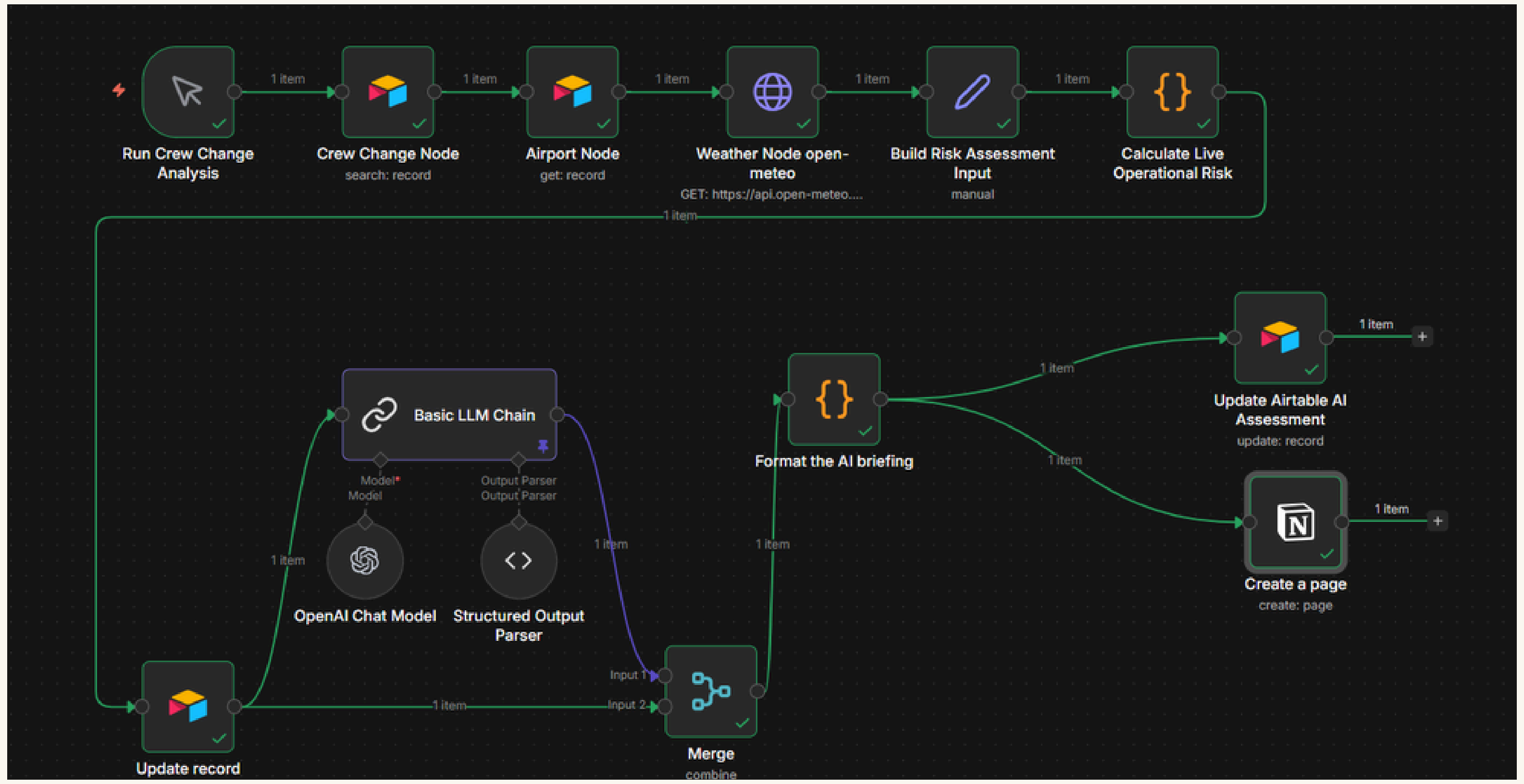
Factors contributing to the operational concern

SUPPORT

Human decision-making, not autonomous decisions

From Concept to Proof of Concept

AI-supported workflow + transparent monitoring



POC WORKFLOW

Crew Change Data

↓

AI Analysis

↓

Risk Briefing

↓

Human Review

✓	Name	Input	Output	Error	Start Time	Latency	Dataset	Annotation ...	Tokens	Cost
✓	Generate AI Risk Briefing	gpt-5-nano	{"briefing":{"ai_recomme...		9/1/2026, 2:07:43 PM	15.64s			2,219	$0.0007...
✓	Generate AI Risk Briefing	gpt-5-nano	{"briefing":{"ai_recomme...		9/1/2026, 2:07:20 PM	22.90s			3,099	$0.0011...
✓	Generate AI Risk Briefing	gpt-5-nano	{"briefing":{"ai_recomme...		9/1/2026, 2:07:01 PM	18.71s			2,569	$0.000...
✓	Generate AI Risk Briefing	gpt-5-nano	{"briefing":{"ai_recomme...		9/1/2026, 2:06:36 PM	25.45s			3,294	$0.0012...
✓	Generate AI Risk Briefing	gpt-5-nano	{"briefing":{"ai_recomme...		9/1/2026, 2:06:14 PM	21.82s			2,517	$0.000...

MONITORING & TRANSPARENCY

Review inputs and outputs • Monitor system behaviour • Investigate unexpected results

Indicative Cost, Timeline & Next Step

A lean, controlled approach to validating the Crew Change Risk Copilot

INVESTMENT

≈ €31,000

Indicative pilot investment

Based on a focused MVP / controlled pilot

TIMELINE

12 WEEKS

Discovery & Planning
Solution Design
MVP Development
UI & Integration
Testing & Refinement
Final Review

NEXT STEP

CONTROLLED PILOT

Validate:
- **Technical feasibility**
- **Operational value**
- **User adoption**

Before larger-scale deployment

Key assumptions:
Focused scope • Pilot data • Limited integrations • Small user group • Human oversight

Crew Change Risk Copilot

AI-supported decision support for smarter and more proactive crew change operations

Use AI to help operations teams identify and understand potential crew-change disruptions — while keeping humans in control.

IDENTIFY

Potential cases requiring review

EXPLAIN

Relevant factors and context

SUPPORT

Human judgement, not autonomous decisions

WHAT I WOULD VALUE YOUR FEEDBACK ON:

Does this use case address a meaningful operational problem and provide a strong enough foundation for a focused MVP?

Thank you

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